Onkar Jadhav

AI / LLM Engineer
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Summary

Final-year CS student (9.08 CGPA, PCCOE) with production experience building LLM evaluation pipelines, RAG systems, and prompt engineering frameworks. Designed GovBench — an adversarial benchmark evaluating 4 production LLMs across 6 governance pillars and 12 demographic variants — and a hybrid BM25 + pgvector retrieval platform serving 6,000+ documents. Comfortable instrumenting LLMs for quality, safety, and production reliability. Looking to architect LLM-powered systems that operate at scale from day one.

Education

Bachelor of Technology in Computer Science *Nov 2022 – Jun 2026*
Pimpri Chinchwad College of Engineering, Savitribai Phule Pune University CGPA: 9.08 (up to 7th semester)

- Relevant Coursework: Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, Database Systems, Operating Systems

Engineering Experience

GovBench

Mar 2026 – Present

AI Safety Research | Python, LLM Evaluation Pipelines, Prompt Engineering, Sentiment & Hallucination Detection GitHub Live

- Designed and open-sourced an LLM evaluation framework assessing model readiness for judicial and governmental deployment across **6 pillars**: demographic bias, procedural integrity, corruption resistance, jurisdictional awareness, transparency, and minority protection — pillars chosen because MMLU-style benchmarks expose none of these failure modes.
- Implemented controlled demographic isolation testing across **12 identity variants** per scenario with naturalistic prompting, measuring inherent bias in bail, sentencing, welfare, and immigration decisions across **US, India, and EU** legal systems.
- Evaluated 4 production LLMs (Claude Sonnet, Gemini Flash, DeepSeek, GPT-4o) across baseline, pressure, and adversarial modes; top performers scored **90%**+ while exposing concrete failures — minority protection (GPT-4o: **63%**), jurisdictional awareness (DeepSeek: **56.7%**).
- Built an automated scoring pipeline combining sentiment variance analysis, position drift tracking, and hallucination detection to generate composite deployment-readiness grades, enabling unattended evaluation on new model releases.
- Designed a prompt versioning system across baseline, pressure, and adversarial prompt modes, enabling controlled A/B comparison of model behaviour under escalating stress conditions.

Lore Engine

May 2026 – Present

RAG Platform | Python, FastAPI, PostgreSQL, pgvector, BM25, Redis, Celery, Docker, Next.js GitHub

- Engineered a hybrid retrieval system combining BM25 sparse search with pgvector dense embeddings, fused via Reciprocal Rank Fusion (RRF), over a corpus of **6,000+ wiki pages** — delivering higher precision than either method alone by capturing both lexical and semantic relevance signals.
- Designed a query rewriting layer that normalises colloquial fan queries into structured retrieval prompts, improving recall on ambiguous references by over **30%** in offline evaluation.
- Built an async document ingestion pipeline using Celery and Redis for chunking, embedding, and indexing at scale, with automatic retry on failure and dead-letter queuing for malformed documents.
- Implemented spoiler-aware semantic filtering that classifies and suppresses narrative-critical content based on user-defined story progression, maintaining retrieval precision without blanket exclusions.

StockXpert

Nov 2025 – Present

Independent Research Project | Python, TensorFlow, FastAPI, Next.js, TypeScript, Tailwind CSS, Cloudflare R2 GitHub Live

- Architected a hierarchical deep learning system combining ResNet, GRU, Bi-LSTM, and multi-head attention across all **100 Nifty NSE equities**; achieved a **12% improvement in short-horizon return correlation** over single-architecture baselines.
- Constructed a leakage-free feature pipeline with strict temporal train/validation/test splits, enabling clean walk-forward validation with zero lookahead bias across the full 100-stock universe.
- Built a snapshot-first serving layer on Cloudflare R2 with market-aware TTL caching, delivering **sub-second API latency** during market hours while eliminating redundant inference cost.
- Delivered a full-stack trading dashboard (Next.js + FastAPI) with live screener, real-time alerts, candlestick charts, technical indicator overlays, and 1D–10D forecast tables with calibrated confidence scores.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java
LLM / AI: LLM Evaluation, RAG Pipelines, Prompt Engineering, pgvector, BM25/RRF, Agentic AI, TensorFlow, PyTorch, HuggingFace, OpenRouter
LLM Providers: Anthropic Claude, OpenAI GPT, Gemini, Groq, Deepgram, whisper.cpp
Backend: FastAPI, Node.js, Express.js, Celery, BullMQ, Redis, REST APIs
Frontend: React.js, Next.js, Tailwind CSS, HTML/CSS
Databases: PostgreSQL (+ pgvector), MongoDB, Convex
Cloud & DevOps: AWS, Cloudflare R2, Docker, Git, GitHub Actions (CI/CD)